VISHAL MENON

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AI Engineer & Data Scientist | LLMs · Agentic AI · Evaluation · PyTorch · Python | Columbia MS Data Science

EDUCATION

Columbia University — New York, NY
***MS in Data Science**, GPA: 3.67/4.0* — Expected Dec 2026
Coursework: Deep Learning, Natural Language Processing, High Performance ML, Probability, Statistics, Product Management for AI

Amrita Vishwa Vidyapeetham — Coimbatore, India
***B.Tech in Computer Science, AI Specialization**, GPA: 9.79/10* — Aug 2023
Silver Medalist (2nd Rank Holder); All-Round Excellence Award; 3 peer-reviewed publications (IEEE, Springer), including plant disease detection with transfer learning

WORK EXPERIENCE

BNP Paribas — New York, NY
Summer Intern, Americas Analytics Lab — Jun 2026 – Aug 2026

- Built an agentic LLM memory-management system using LangChain, LangGraph, CrewAI, and AutoGen with a 20-member Americas Analytics Lab team, evaluating knowledge bases across 50+ documents for long-context reasoning in financial services.
- Constructed an LLM memory evaluation framework achieving ~0.65 F1 on evidence retrieval and multi-hop QA tasks, benchmarking 3+ open-source LLMs and memory architectures against a RAG baseline for scalable Agentic AI.
- Evaluated 6+ dimensions of LLM memory quality including injection, retrieval, updates, duplication, and knowledge organization, defining metrics for production-grade memory systems in financial AI workflows.
- Presented results to technical and business stakeholders, translating evaluation metrics into plain-language findings.

JPMorgan Chase & Co — Bengaluru, India
Software Engineer-I, Corporate Technology — Jun 2023 – Jun 2025

- Led the team as Project Manager during a major release cycle, driving stakeholder alignment and feature prioritization; received the Impact Award 2023 for ML monitoring solutions adopted across corporate technology teams.
- Formulated evaluation metrics for 70% faster model deviation and drift detection in production, enabling real-time monitoring across 10+ finance ML use cases, including market risk models, and faster incident response.
- Innovated 15 LLM evaluation metrics spanning prompt quality, response relevance, hallucination checks, and RAG assessment; implemented an evaluation framework across 70+ production prompts and response chains.
- Developed an LLM-based text-to-SQL system that generates SQL queries from natural-language questions.

Attachment Intern — Feb 2023 – May 2023

- Implemented an ML monitoring website with explainability, drift, and bias checks across 10 production finance use cases, creating an observability framework that achieved 3x evaluation efficiency.

RESEARCH & PROJECTS

Uncertainty-Aware LLM Inference (IBM Research mentored) — Columbia University
Python, PyTorch, vLLM, Quantization, CUDA Profiling (Nsight), W&B, A100 GPUs — Jan 2026 – Aug 2026

- Co-authored a paper on post-training quantization and LLM confidence calibration across 5 configs (FP16, GPTQ INT4/INT8, AWQ INT4, NF4) and 3 datasets, measuring ECE, MCE, Brier score, and entropy.
- Showed GPTQ INT4 delivers 1.78x vLLM throughput vs FP16 with 70% less GPU memory (4.3 GB vs 14.5 GB), profiling kernels with Roofline analysis.
- Simulated confidence-based model routing that cut inference cost by 69.8% at equal or better accuracy on PubMedQA.

NLP and LLM Analysis of Psychological Health Records — Columbia University
Python, NLP, LLMs, Clinical NLP, Knowledge Graphs — Jan 2026 – Aug 2026

- Applied NLP and LLMs to clean and analyze noisy NVDRS narratives from 50k+ reports, building domain-specific text processing pipelines with privacy compliance.
- Collaborated with psychologists and clinicians to interpret domain terminology, translating clinical concepts into structured data and a knowledge graph for relationship extraction.

SKILLS

LLMs & Agentic AI: LLM evaluation, RAG, agentic AI, LangChain, LangGraph, CrewAI, AutoGen, LlamaIndex, FAISS, LoRA/QLoRA fine-tuning, prompt engineering, hallucination detection, text-to-SQL

ML, Statistics & Inference: PyTorch, TensorFlow, Keras, scikit-learn, transfer learning, CNNs, vision transformers, time series (ARMA), Bayesian modeling, model calibration, vLLM, quantization (GPTQ, AWQ), CUDA profiling

Languages, ML Systems & Cloud: Python, SQL, Java, R, drift detection, production monitoring, observability, SHAP, LIME, AWS, GCP, Kubernetes, Docker, Terraform, Spark, Git

Leadership & Communication: Team leadership, project management, stakeholder management, cross-functional collaboration, technical presentations, Agile

LEADERSHIP & TEACHING

President, Data Science Institute Student Club (DSISC) — Columbia University | Current

- Lead the executive board, coordinating events and initiatives with students, faculty, and industry partners.

Teaching Assistant, Data Analytics and Machine Learning (IEORE4523) — Columbia University | Current

- Support a graduate analytics and ML course, holding office hours and explaining ML concepts to 300+ students.